TD of the simulation of CO2 emissions of 4 cars on specific driving cycles (ICE / BEV / HEV / PHEV)

TD description: (work by group of 3 persons)

TD inspired with authorization by the computer project in automotive engineering, Prof. P. Duysinx (ULiege) and M. Nélis (Certificate)

We ask to consider vehicles equipped with petrol engine, hybrid engine and electric motor in order to compare them. These vehicles are known by their technical characteristics: size, mass, power, torque, wheel size, engine bmep, bsfc, efficiency models...

It is asked to consider a specific travel that these 3 vehicles will perform. This travel is known by the distance and the elevation of the road. The speed profile of the vehicle will be determined from the performance analysis and cornering capabilities (vehicle dynamics).

You need to discretise this travel to calculate the state and the energy consumption of the vehicles at each point or during each segment.

It is required to make a digital trajectory of the vehicles by means of physics formulas to evaluate at each point of the travel the power used by the powertrain and thus estimate the vehicle energy consumption and its CO2 emissions.

The values finally obtained will be reduced to specific quantities namely l/100km or kwh/100 km for energy consumption and gr/km for CO2 emissions.

It is finally asked to analyse the results obtained by comparing vehicles in terms of consumption and emissions. A performance-based comparison is also possible.

It is asked to identify the weaknesses of the modelling and to criticise/improve it. What is the next step?

The main goal of this work is to generate a ppt presentation with a short introduction made of the general simulation approach which will be completed by each student of the group by a personal study of the influence of a relevant parameter that the existing modelling could perform, eventually with minor adaptations. This added study will be your personal contribution and will be complementary to the work of the group.

- Influence of a relevant parameter (not the gravity or include all effects)
- A technical description linked to emissions, energy storage…
- A detailed simulation in vehicle dynamics like cornering, aero

1. Theories needed for the calculation of vehicle performance

- Simulation of a specific travel by time-distance discretisation
- Tire behaviour, low speed cornering
- Radial acceleration and maximum speed when cornering
- Force due to longitudinal acceleration
- Force due to rolling resistance
- Force due to aerodynamic resistance (drag force)
- Force due to grading
- Power needed at the wheels
- Transmission efficiency
- Power delivered by the engine
- Energy consumed by the engine
- Conversion between energy and fuel consumption
- Conversion between fuel consumption and CO2 emissions

2. Vehicles characteristics

- 1 petrol car, 1 electric vehicle, 1 hybrid car, 1 plug-in hybrid
- Mass of the vehicle (kg), mass of the battery (kg)
- Engine power curve vs rpm - P(N)
- Engine torque curve vs rpm - C(N)
- Transmission ratio(s)
- Tires (size, profile, cornering stiffness)
- Frontal surface S ; Drag coefficient Cx
- Transmission efficiency
- BSFC mapping of the engine vs (BMEP, RPM)
- Estimation of engine efficiency vs power – η(P)
- Mapping of electric motor efficiency vs (Torque, RPM)
- Estimation of motor efficiency vs RPM

3. Travel considered

To be presented for information (NEDC driving cycle by distances, gear engaged and car velocity profile)

- Francorchamps race track defined by the trajectory (distances, altitudes, corners) and the speed profile at different points as follows:

  Case number 1, soft eco driving: (same for all cars)
  - Speed in corners limited by the tire friction coefficient
  - Consider rapid corners 16 and 17 like straight lines
  - Longitudinal acceleration: 0-100 kph in 20 s
  - Longitudinal deceleration: ~0,4g
  - Maximum speed of 90 kph

- Engine power will probably be enough

Case number 2, sports driving: (same for all cars)

- Speed in corners limited by the tire friction coefficient, good trajectory
- Consider rapid corners 16 and 17 like straight lines
- Longitudinal acceleration: depending of IC engine capability
- Longitudinal deceleration: depending on tire friction coefficient (~0.6g)
- Maximum speed imposed by the IC engine power
- Iterations needed to find the max speed point and braking point!

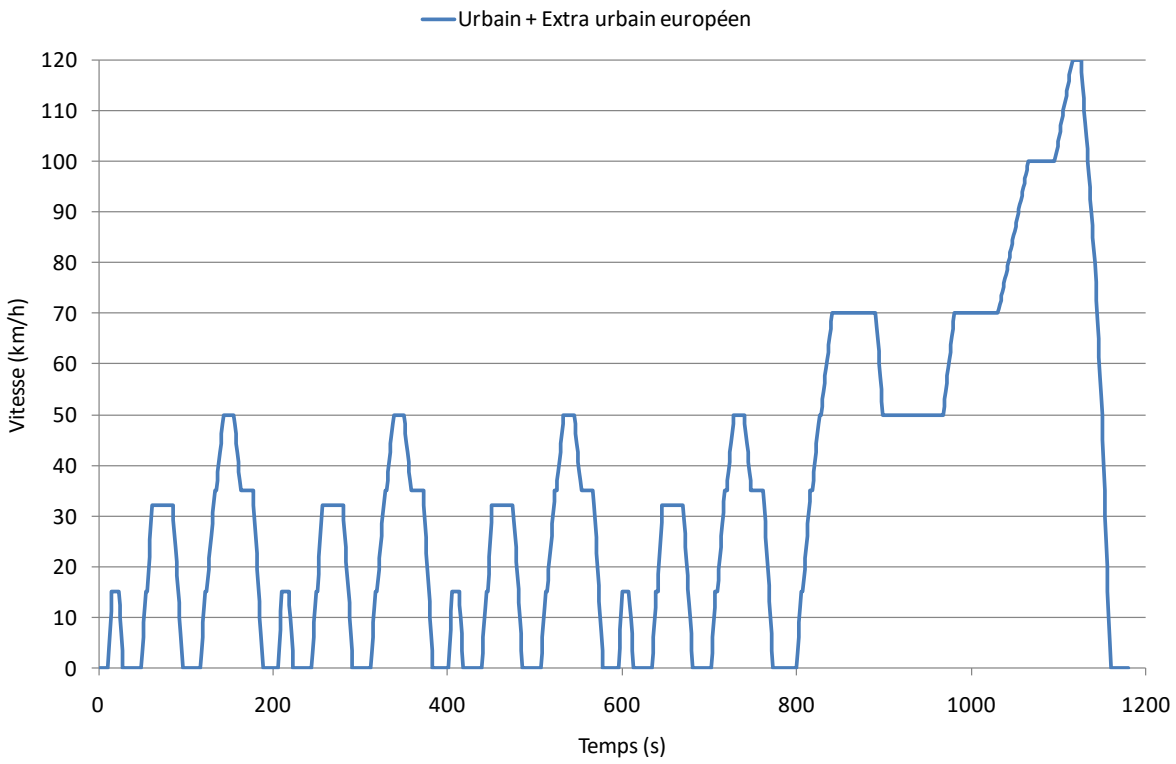
4. Recommended steps for this simulation project

Discretize the trajectory in "n" points (Excel sheet for me) and for each point you must calculate/estimate:

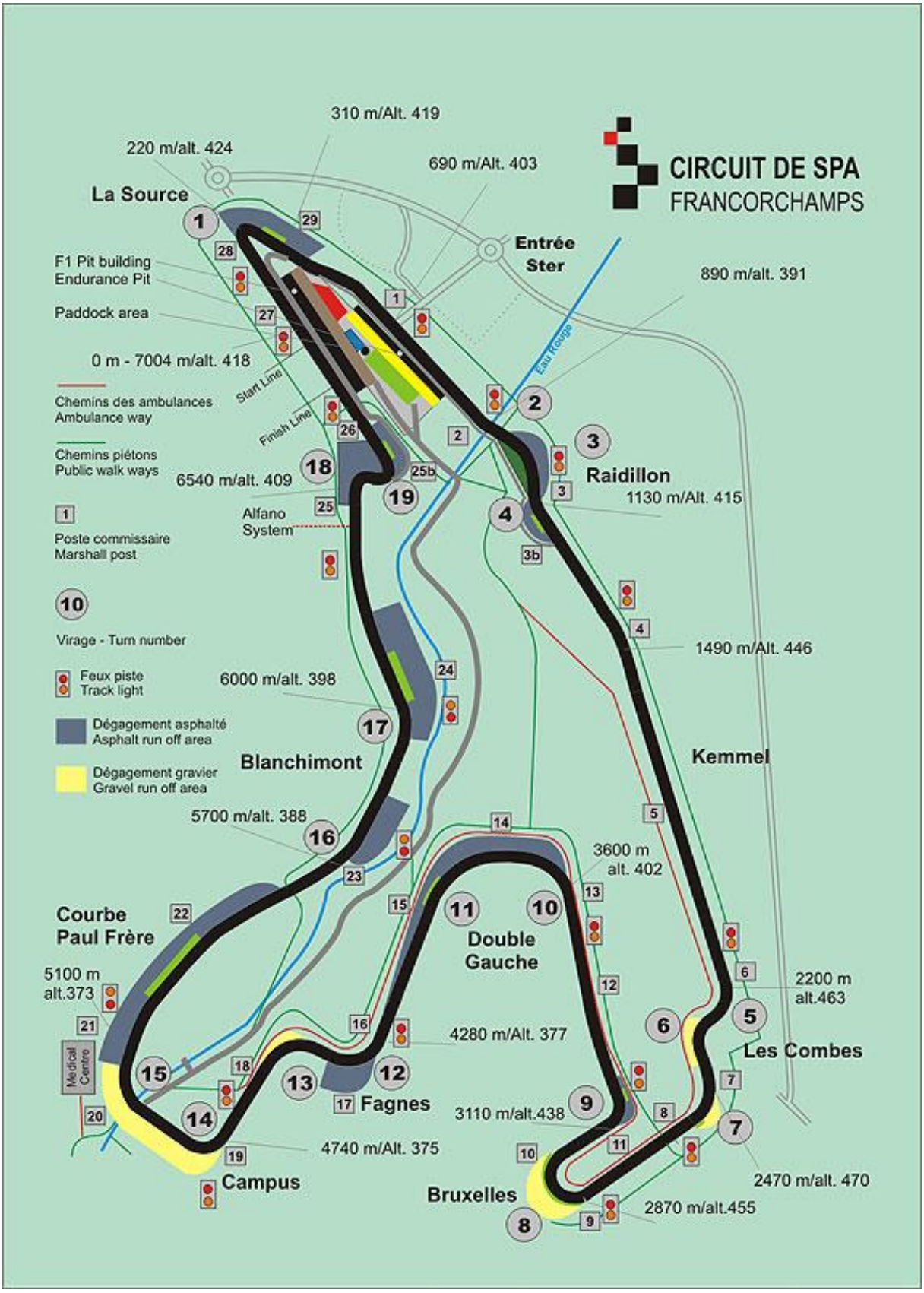
- Distance since the start point
- The altitude of the point
- The vehicle speed
- The vehicle longitudinal acceleration
- All resistance forces: acceleration (inertia), drag, rolling friction, grading
- The net force needed to maintain the speed (the sum of all resistance forces)
- (The torque at the wheels)
- The transmission ratio engaged vs car speed
- (The engine torque)
- The power at the wheels
- The engine power at engine (including transmission efficiency) and check capability
- The fuel consumption (introducing engine efficiency and LHV of the fuel)
- CO2 emissions in grams (conversion formulae)
- Integration of the results on the complete driving cycle
- Convert results in l/100 km or kWh/100 km for the fuel consumption and in g/km for CO2 emissions

DATA

1. NEDC cycle (for information)



2. Francorchamps race track: trajectory, distances and altitudes



+ Video of the track

3. Data of the petrol car

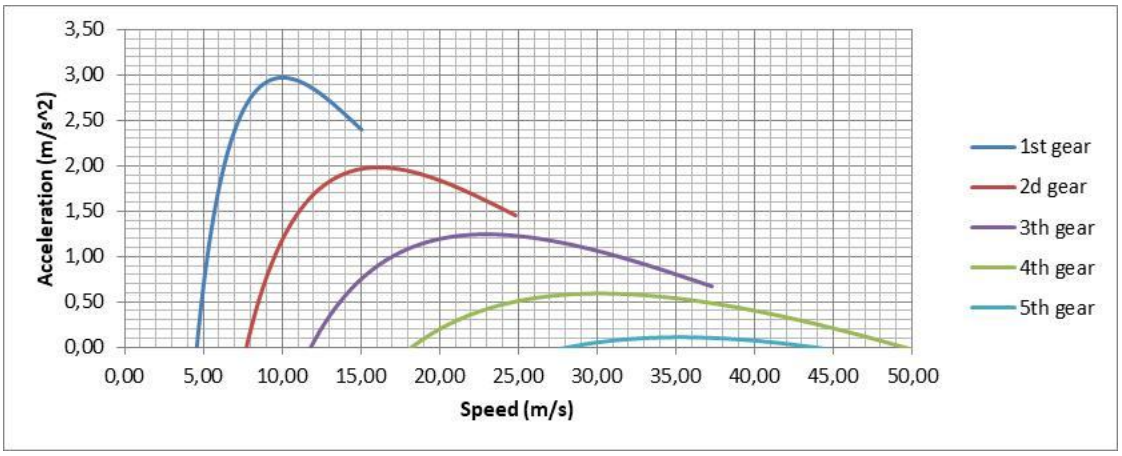
Type COMPACT gasoline 1.6i naturally aspirated

Mass	1280 kg
Frontal surface	2.69 m^2
C_x	0.31
Engine power MAX	74 kW à 6000 rpm (100CV)
Engine torque MAX	146 Nm à 4000 rpm
Transmission ratio	i1 = 12,526
	i2 = 7.588
	i3 = 5.052
	i4 = 3.428
	i5 = 2.525
Power curve	Linear between 0 and 5000 rpm, plateau between 5000 and 6000 rpm
Torque curve	to be calculated
Transmission efficiency	95%
Engine efficiency	20% to 40% (NA to downsized)

(The main influence comes from the power, in the simulation do not put 0% at 0 power!)

Tires	195-65-R15
	(External diameter of the tire 634.5mm)

Choice of the engaged gear speed:



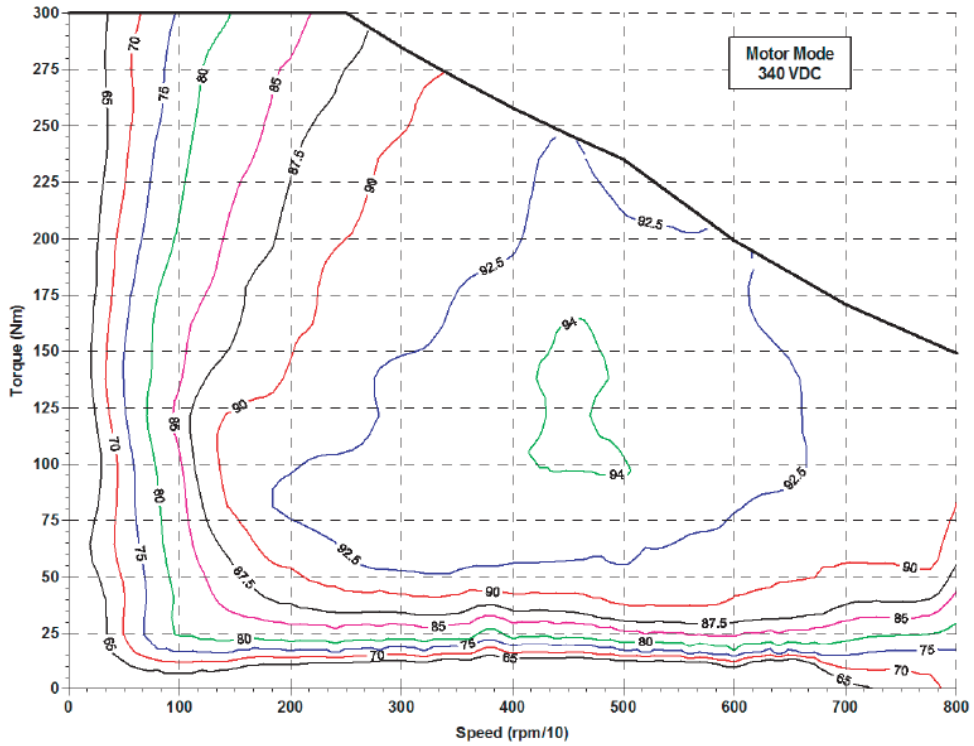
Approximately 36, 57, 82, 108 kph - will be calibrated in the model

4. Data for the electric vehicle

Type Generic electric basics	
Mass (including 22 kWh batteries)	1460 kg (1230 kg car only)
Frontal surface	2.69 m^2
C_x	0.31
Electric motor peak power	125 kW à 8000 rpm
Electric motor nominal power	45 kW à 3000 rpm
Electric motor torque (peak)	300 Nm
Electric motor torque (nominal)	150 Nm
Transmission ratio	6.218
Electric motor MAX efficiency	94%
Efficiency of the transmission	95%
Tires	195 65 R15 (External diameter of the tire 634.5mm)

Mapping torque-efficiency-speed of the electric motor

The main influence comes from the speed !



5. Data for the hybrid vehicle

We will use the excel file elaborated for the petrol car and adapt it to this vehicle!

- Same car than the petrol car
- Same internal combustion engine than the petrol car

Mass increased of:

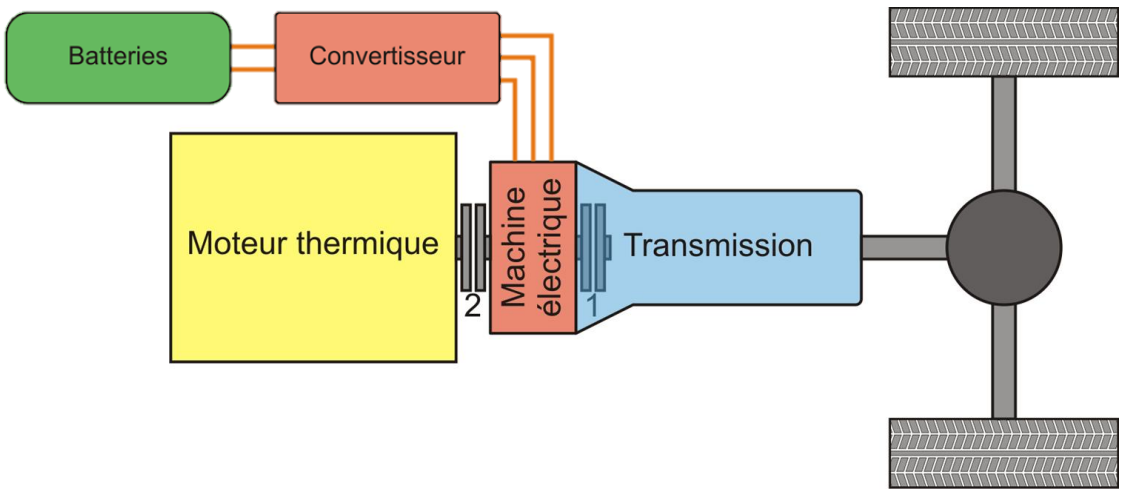
- Electric machine and inverter (100 kg)

- Battery 150 Wh/kg (HEV 1.5 kWh or PHEV 10-20 kWh)

Electric power:

HEV hybrid : 20-30 kW – Torque : 100 Nm

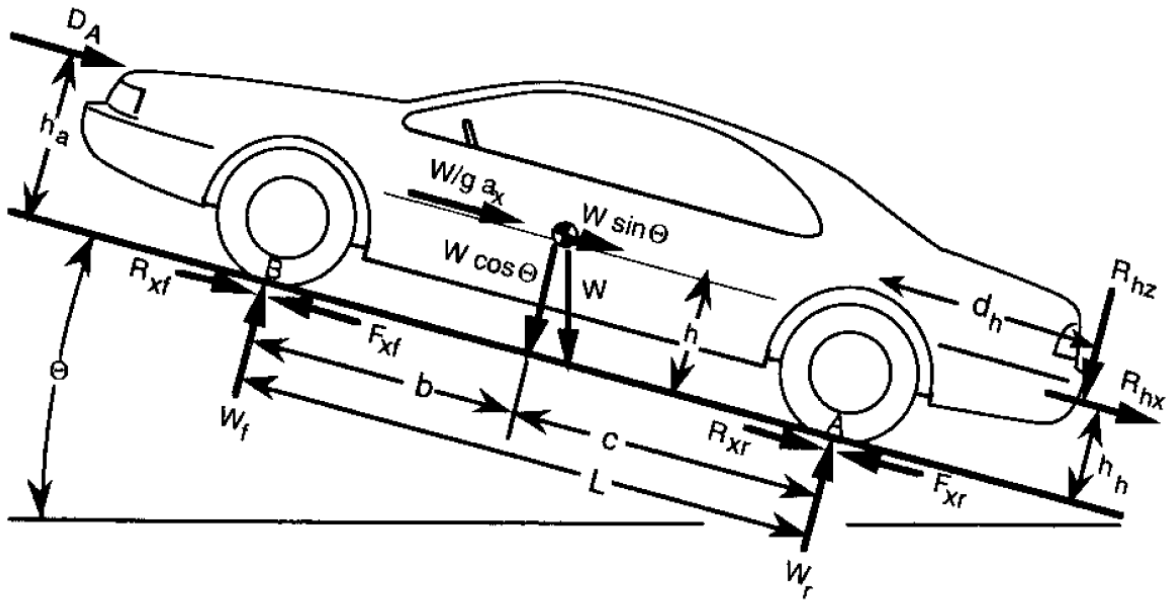
PHEV Plug in hybrid : 40-50 kW – Torque : 200 Nm



6. Theory – useful formulas

Reference 1: Cours « Performance des véhicules » Prof. P. Duysinx, ULiege

Reference 2: fundamentals of vehicle dynamics, T. Gillespie



6.1 Force due to mass inertia and transmission inertia

- Estimation of acceleration and elasticity is based the second Newton law

$$F_w - \sum F_{res} = F_{net} = m\frac{dV}{dt}$$

- Warning: when accelerating, one is also increasing the rotation speed of all driveline and transmission components: wheel sets, transmission shafts, gear boxes and differential, engine…

➔ Effective mass to account for the kinetic energy of all components (translation + rotation)

- One defines an effective mass $T = 1/2\, m_e\, v^2$

$$m_e = m + \frac{\sum I_w + I_{shaft}}{R_e^2} + \frac{(I_{transm} + I_{box2})\, i_{dif}^2}{R_e^2} + \frac{(I_{box0})\, i_{dif}^2 i_{box}^2}{i_{box0}^2\, R_e^2} + \frac{(I_{box1} + I_{clutch} + I_{cs})\, i_{dif}^2\, i_{box}^2}{R_e^2}$$

- The calculation of the effective mass requires the knowledge of the geometry of all the driveline components

- Empirical formula for preliminary design of cars by Wong

$$\gamma_m = \frac{m_e}{m} = 1.04 + 0.0025\, i^2 \qquad i = i_{dif} * i_{box}$$

6.2 Force due to rolling resistance

- Experiments show that generally, the global rolling resistance forces are, with a very good agreement, using a linear model as a function of the vertical force applied onto the tire

$$\boxed{F_{rl} = f\, W = f\, m\, g}$$

The coefficient *f* is the rolling resistance coefficient

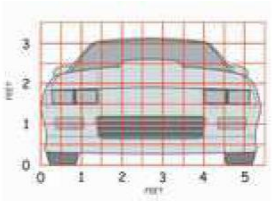
- **The rolling resistance coefficient,** ratio between the rolling resistance force and the normal force encompasses the complicated and interdependent physical properties of the tire and the ground.

Road surface	Coefficient of rolling resistance *f*
Pneumatic car tires on	
Large sett pavement	0.015
Small sett pavement	0.015
Concrete, asphalt	0.013
Rolled gravel	0.02
Tarmacadam	0.025
Unpaved road	0.05
Field	0.1...0.35
Pneumatic truck tires on	
concrete, asphalt	0.006...0.01
Strake wheels in field	0.14...0.24
Track-type tractor	
in field	0.07...0.12
Wheel on rail	0.001...0.002

6.3 Force due to aerodynamic resistance

- Drag force

$$F_{\text{aéro}} = \tfrac{1}{2}\,\rho\, S C_x\, V^2$$



- Estimating the frontal area
  - Paul Frere formula

$$S = k\, h\, l \quad \text{avec} \quad k \approx 0.85$$

  - Wong formula

$$S = 1.6 + 0.00056\,(m - 765) \quad [m^2]$$

6.4 Force due to grading

From the diagram we consider - m g sin θ

As a summary we finally consider:

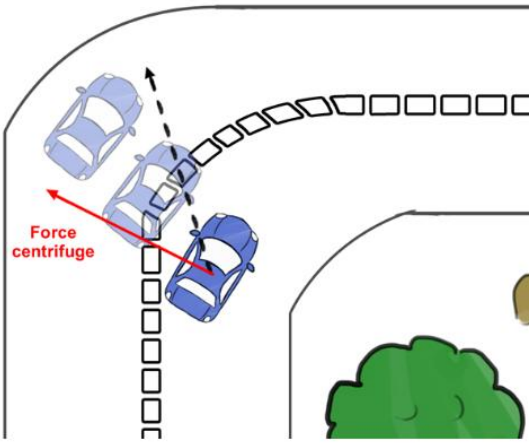
- Equilibrium along forward x direction

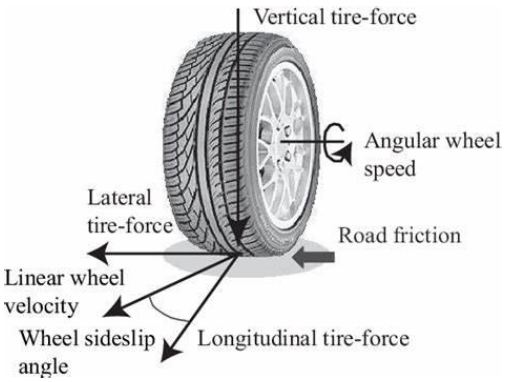
$$ma_x = \sum_i F_{xi} - \sum_i R_{xi} - mg\sin\theta - D_{Aero}$$

- Equation of vehicle longitudinal motion

$$ma_x = F_x - R_{rl} - mg\sin\theta - D_A - R_{h,x}$$

6.5 Radial acceleration when cornering



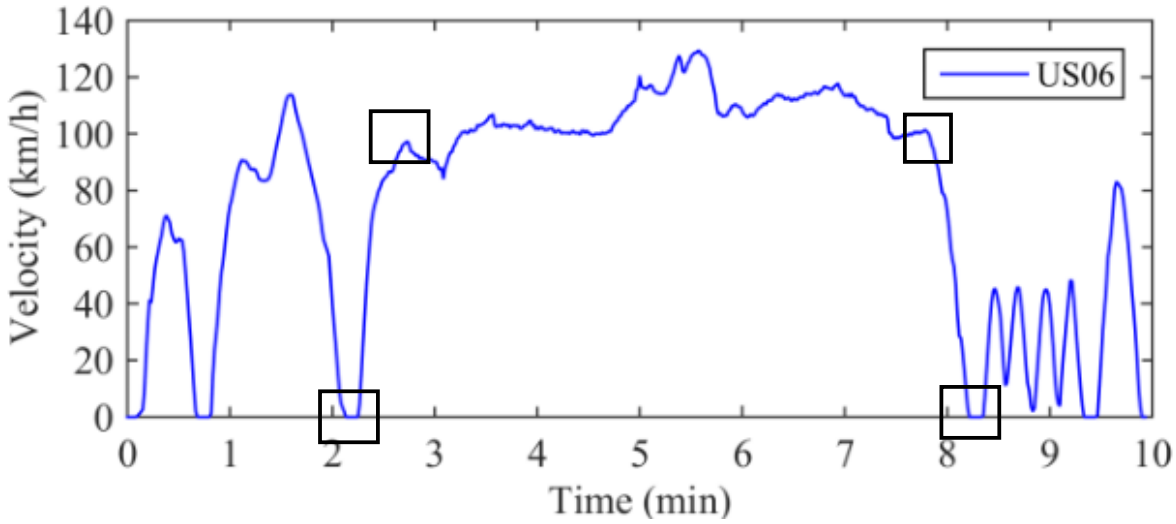


Radial acceleration=V^2/R ; V vehicle speed and R radius of the corner

6.6 Power at the wheels

Net force that makes the car moving * vehicle speed

!! After braking the vehicle speed is zero and the power recovery also !!



6.7 Efficiency of the transmission

This efficiency is evaluated by measuring friction losses in gears; it is often over 90%

6.8 Power delivered by the engine

Power at the wheels divided by the transmission efficiency ; or multiplied by efficiency during braking

6.9 Conversion between power and fuel consumption

Do not forget to include the engine efficiency and then use the LHV of the fuel and the fuel density. You can also use the engine fuel consumption mapping bsfc when available.

1 kg gasoline or diesel ~ 40 MJ or 11 kWh ; gasoline density = 0.75

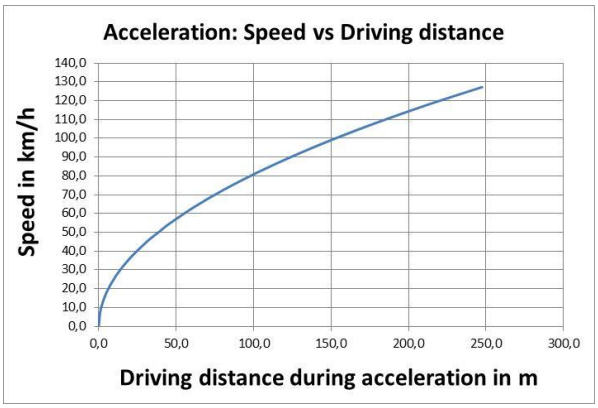
6.10 Conversion fuel consumption into CO2

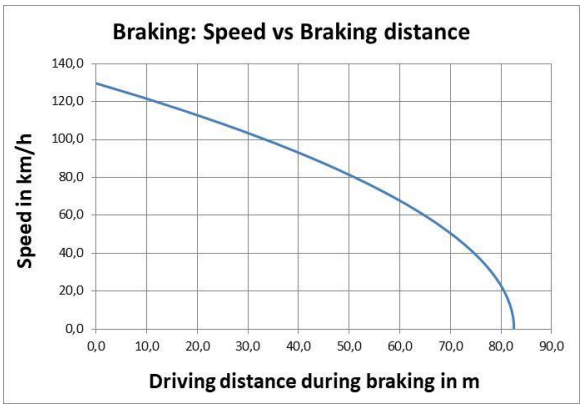
$$C_aH_b + \left(a + \frac{b}{4}\right)(O_2 + 3.773\,N_2) = aCO_2 + \frac{b}{2}\,H_2O + 3.773\left(a + \frac{b}{4}\right)N_2$$

1 litre diesel	~	2640 g CO2
1 litre gasoline	~	2392 g CO2
1 litre LPG	~	1665 g CO2
1 kg CNG	~	2536 g CO2
1 kWh elec	~	250 g CO2 (European average)
1 kWh elec	~	73 g CO2 (France)

Source: https://www.eea.europa.eu/data-and-maps

6.11 Calculations needed to build the velocity profile on the track lap





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END of the presentation of the TD

Some references:

Car specifications
https://www.automobile-catalog.com/#gsc.tab=0

WLTC driving cycle
https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Funece.org%2Ffileadmin%2FDAM%2Ftrans%2Fdoc%2F2012%2Fwp29grpe%2FWLTP-DHC-12-07e.xls&wdOrigin=BROWSELINK

SSP VAG Self-study program (Many temporary links on the web)
https://www.vdveer-engineering.nl/en/information/vag-ssp/ssp-vag/ssp-vag-fr
https://www.vaglinks.com/Docs/SSP/